

Problem: Smart Home Device Interface

You are developing software for a smart home system. Multiple types of smart devices (like lights, thermostats, and security cameras) need to follow a standard interface so the central controller can interact with them uniformly.

Task:

1. Define a SmartDevice interface with the following methods:
 - o void turnOn()
 - o void turnOff()
 - o boolean isOn()
2. Implement the SmartDevice interface in the following classes:
 - o SmartLight: Has an additional method void setBrightness(int level) (0-100).
 - o SmartThermostat: Has an additional method void setTemperature(double temperature).
3. Write a class SmartHomeController that has a list of SmartDevice objects and:
 - o Can add devices using void addDevice(SmartDevice device)
 - o Has a method void turnAllOn() to turn on all devices
 - o Has a method void showDeviceStatus() to print whether each device is on or off (use instanceof to print extra info, like brightness or temperature, if applicable)